Geospatial Site Screening Report

Orojo Resources, Vichada, Colombia | Prepared 18 March 2026 | Screening layers T1 through T6

Site Overview: Orojo Resources, Vichada, Colombia

Native Llanos savanna site (100% persistent non-forest, 1960-2019). Primary finding: the parcel directly overlaps a mapped Resguardo Indigena, triggering FPIC under ILO Convention 169 / Colombian Law 21/1991 (T6 RED).

PROJECT AREA
111 km²

CENTROID
lon -67.7387, lat 6.0151

REALM (RESOLVE)
Neotropic

DOMINANT BIOME
Tropical & Subtropical Grasslands, Savannas & Shrublands

PERSISTENT NON-FOREST (ESTOQUE ET AL.)
100.0%

GLC_FCS30D (30M, MEAN 1985-2022)
89.3% grassland / 4.3% forest GREEN

NCS REFORESTATION (GRISCOM)
0.0% FLAG (expected biome)

ALBEDO OFFSET MEDIAN (HASLER)
+0.7% POSITIVE

IPLC PROXIMITY (LANDMARK)
RED

Screening Flags (5 total)

- NOTE: Savanna/grassland baseline confirmed: 100.0% persistent non-forest
- NOTE: GLC_FCS30D confirms 89.3% mean grassland cover (1985-2022); forest signal (4.3%) consistent with stable gallery forest
- CONTEXT: NCS overlap 0.0% in Tropical & Subtropical Grasslands, Savannas & Shrublands: low overlap is expected; Griscom explicitly excludes this biome type for biodiversity reasons (Veldman et al. 2015)
- POSITIVE: Median albedo offset 0.7%: net cooling co-benefit
- RED: T6 IPLC: direct overlap with 1 territory, FPIC triggered

T1 - Project Area

The project boundary is loaded from the supplied KML file and used as-is, with no rectangular bounding box approximation is applied. Area is computed in the EPSG:6933 World Equal-Area projection to avoid the distortion that comes from measuring area directly in degrees at tropical latitudes. The geometry is validated as a simple, non-self-intersecting polygon before any downstream analysis. All raster layers (T3-T5) are masked to this exact polygon shape.

Results

Parameter	Value
Boundary source	KML polygon (Polygon)
Polygon envelope (WGS84)	minx -67.79949, miny 5.93812, maxx -67.66234, maxy 6.09554
Area computed in EPSG:6933	110.9 km²
Centroid	lon -67.7387, lat 6.0151
Geometry validity	Valid polygon, no self-intersections
Output CRS	EPSG:4326 (WGS84)

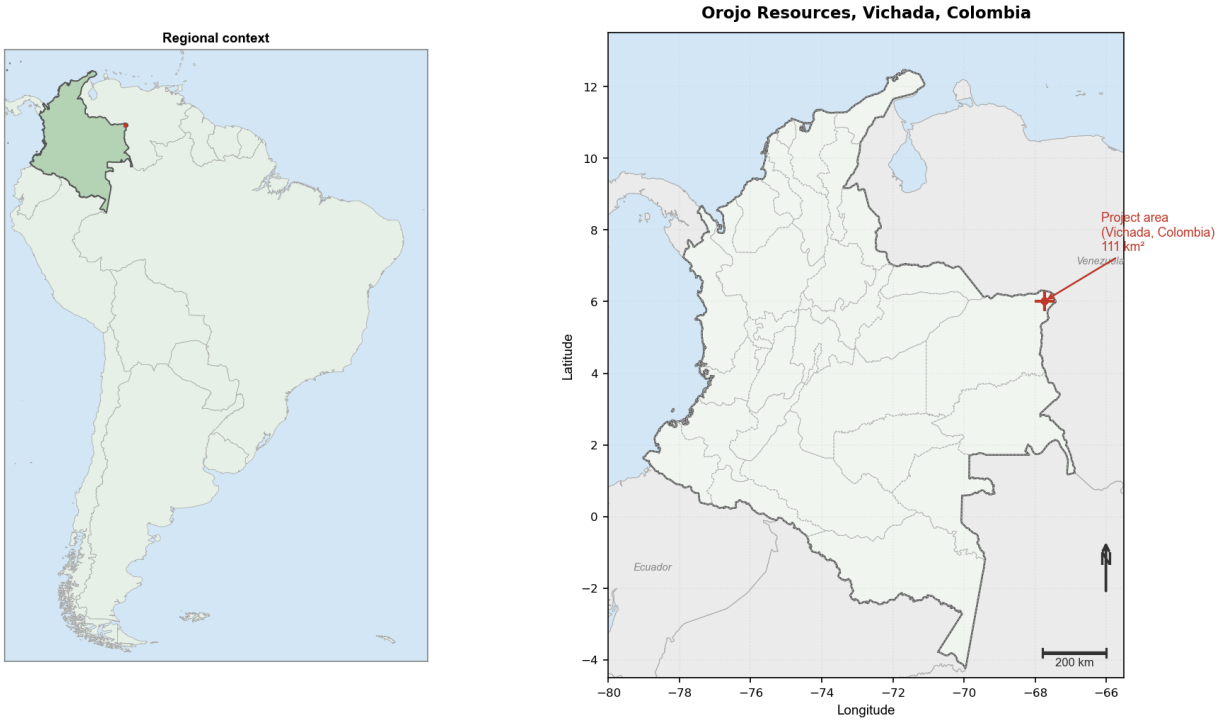


Figure 1: Regional context map. Left panel: South America locator showing project position within the continent. Right panel: Colombia with department boundaries, project bounding box, north arrow, and scale bar.

T2 - Ecological Identity (RESOLVE Ecoregions, Dinerstein et al. 2017)

The project boundary is intersected with the RESOLVE 2017 terrestrial ecoregion dataset to establish native vegetation context and biogeographic realm. Each intersecting ecoregion is weighted by its intersection area. Biome percentages are the sum of all ecoregion areas sharing the same biome classification. Detailed assumptions and implementation choices are in the Key Assumptions table below.

Key Assumptions

Author Assumptions (Dinerstein et al. 2017)	Our Implementation Choices
847 terrestrial ecoregions defined by species distribution data rather than political boundaries. Boundaries reflect long-term biogeographic patterns established over geological timescales.	The RESOLVE shapefile is used in its native EPSG:4326 projection. Intersection areas are computed after reprojecting to EPSG:6933 (equal-area) to ensure accurate km² values. Percentages are derived from these equal-area intersection results.
The 14-biome and 8-realm classification is a static representation. It reflects historical vegetation patterns, not current land use or recent degradation. An ecoregion classified as tropical moist broadleaf forest may contain heavily degraded or converted land.	RESOLVE biome assignment is used as a proxy for native vegetation context, not current state. Current forest cover trajectories are assessed independently in T3 using Estoque et al. 2022. The two layers are complementary and should be read together. A dedicated LULC layer will be added in a later iteration to assess current conditions directly.
Ecoregion boundaries are drawn at a global scale. At sub-regional scales, the boundaries carry inherent positional uncertainty, and small percentages near edges should be treated cautiously.	The dominant biome is determined by the largest intersection area. Ecoregions contributing less than 1% of area are reported but given low interpretive weight.

Guardrails

Check	Threshold	Result	Status
Ecoregion coverage sums to 100%	Within 1% of 100%	100.00%	PASS
No negative intersection areas (geometry artifact check)	Zero negative values	0 negative area polygons	PASS

Results

Dominant biome: Tropical & Subtropical Grasslands, Savannas & Shrublands (100.0%) **Realm:** Neotropic (100.0%)

Ecoregion	Biome	Area (km²)	Coverage
Llanos	Tropical & Subtropical Grasslands, Savannas & Shrublands	111	100.0%

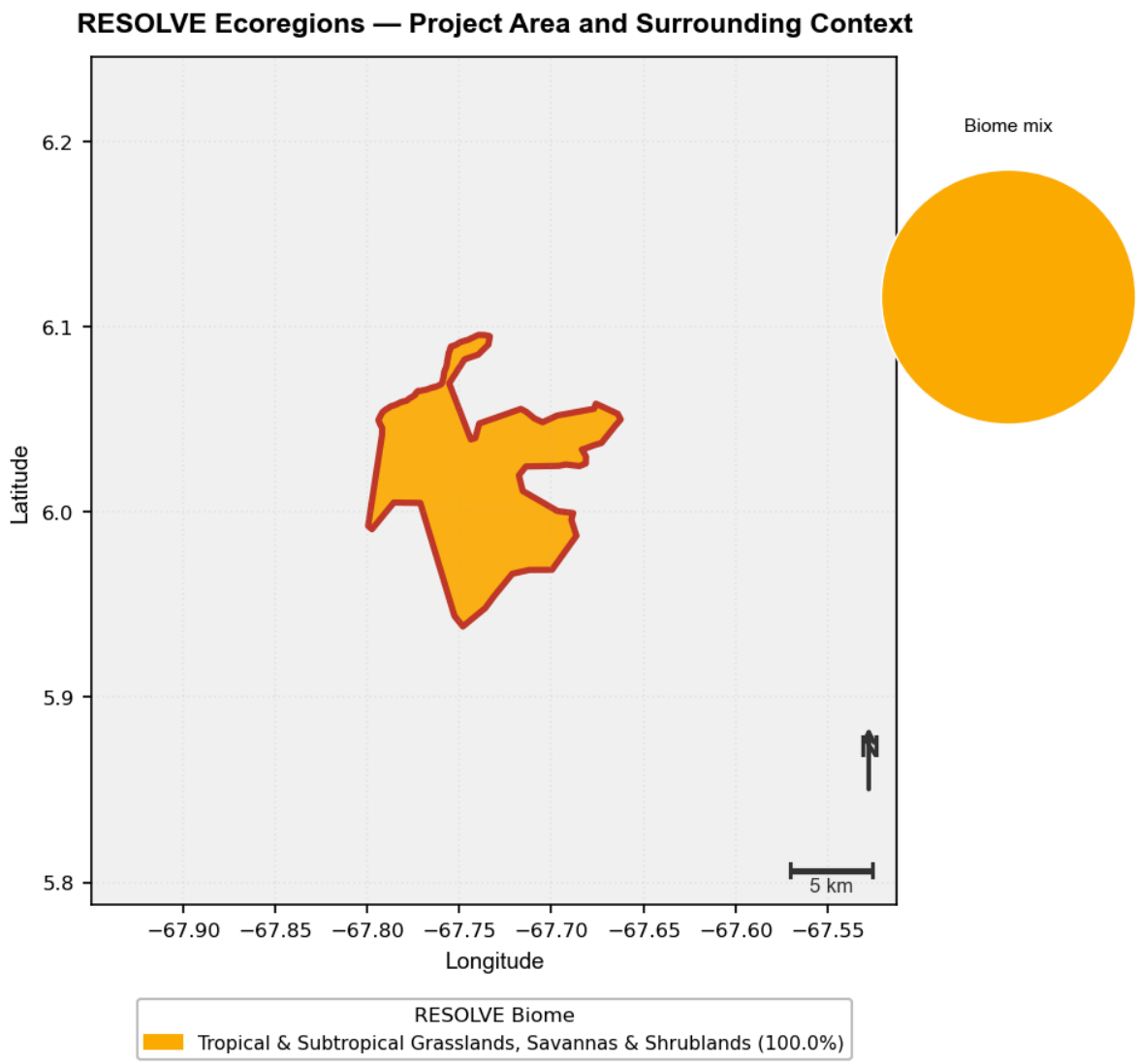


Figure 1: RESOLVE ecoregion overlay. Project boundary intersected with RESOLVE Ecoregions 2017 (Dinerstein et al. 2017). Biome colours follow the RESOLVE standard; biome and realm assignments computed in EPSG:6933 equal-area projection.

Reference Domain Context

The site is classified within the **Llanos** ecoregion (Tropical & Subtropical Grasslands, Savannas & Shrublands biome, Neotropic realm). Two reference domains are used for additionality baseline and leakage belt analysis (RESOLVE Ecoregions 2017, Dinerstein et al.):

- **Domain A - Ecoregion anchor:** All Llanos polygons. Highest comparability, same floristic composition and land cover dynamics as the project site (Figure 2, left panel).
- **Domain B - Biome anchor:** All 11 Neotropic Tropical & Subtropical Grasslands, Savannas & Shrublands ecoregions. Llanos is the closest analogue; 3 of 11 reach HIGH comparability. Full ranking in Table 1 (Figure 2, right panel; Figure 3).

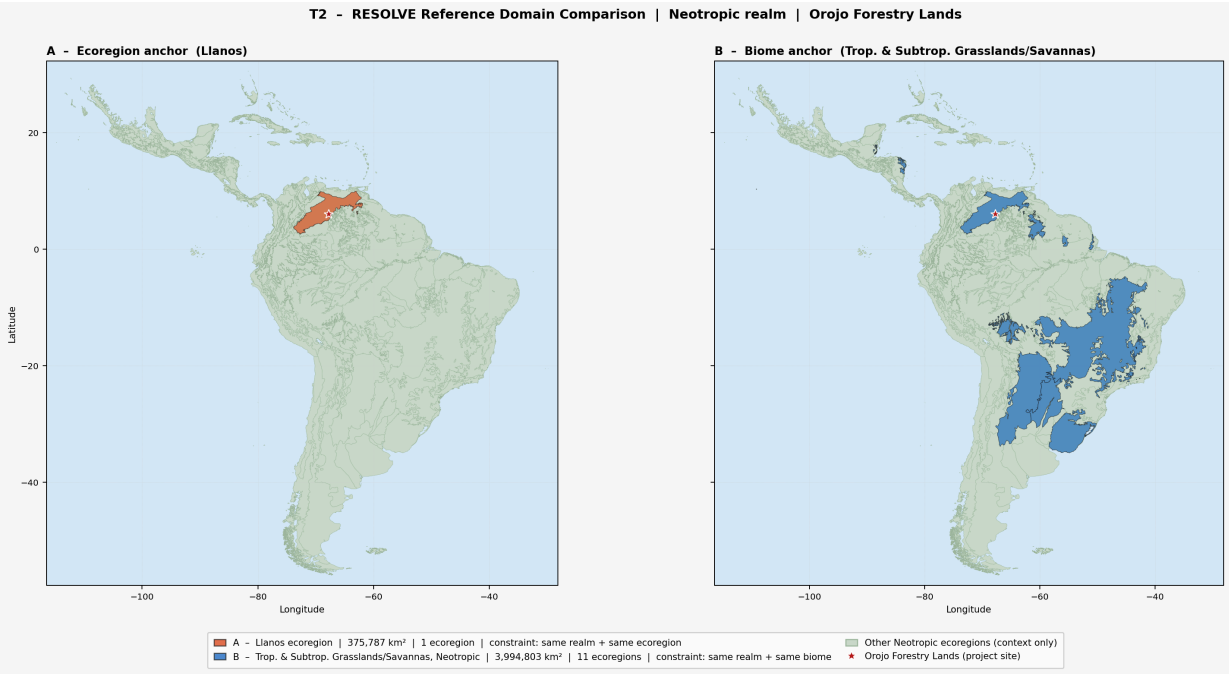


Figure 2: Domain A vs B, Neotropic realm. Left: Llanos ecoregion (Domain A). Right: Tropical & Subtropical Grasslands, Savannas & Shrublands biome (Domain B). ★ = project site.

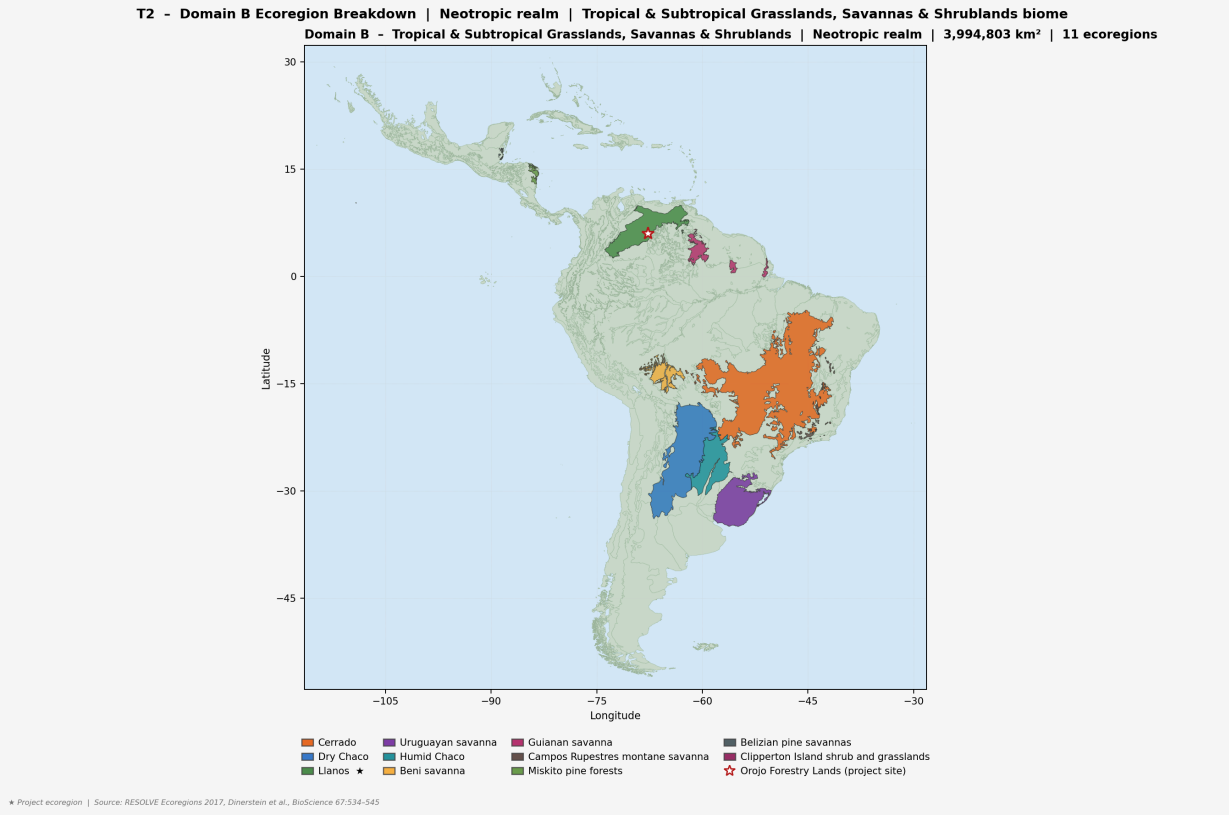


Figure 3: Domain B ecoregion map. Each colour represents one ecoregion within the Tropical & Subtropical Grasslands, Savannas & Shrublands biome, Neotropic realm. Colour key and rankings in Table 1. ★ = project site.

Table 1: Domain B ecoregion ranking. Comparability to Llanos (project anchor): HIGH = lowland tropical savanna, same latitude band; MEDIUM = tropical/subtropical, distinct moisture or geography; LOW = extratropical or ecologically distinct. ★ = project site. Source: RESOLVE Ecoregions 2017.

Ecoregion	Comparability	Share of Domain B	Area
Cerrado	MEDIUM	47.8%	1.9M km²
Dry Chaco	LOW	19.7%	787k km²
Llanos ★	HIGH	**9.4%**	**376k km²**
Uruguayan savanna	LOW	8.8%	352k km²
Humid Chaco	MEDIUM	7.3%	292k km²
Beni savanna	HIGH	3.1%	126k km²
Guianan savanna	HIGH	2.6%	104k km²
Campos Rupestres montane savanna	LOW	0.7%	26k km²
Miskito pine forests	LOW	0.5%	19k km²
Belizian pine savannas	LOW	0.1%	3k km²
Clipperton Island shrub and grasslands	LOW	0.0%	29 km²

T3 - Forest Cover Trajectories, 1960 to 2019 (Estoque et al. 2022, HILDA+)

This layer is derived from two sources: HILDA+ (Winkler et al. 2021), a global land use change dataset at 1 km resolution, processed and classified by Estoque et al. 2022. HILDA+ records land cover state at 7 decadal snapshots: 1960, 1970, 1980, 1990, 2000, 2010, and 2019. Each 1 km pixel is assigned a trajectory label based on how its forest/non-forest state changed (or did not change) across all 7 snapshots.

What "persistent" and "non-persistent" mean: read this before interpreting results.

- **Persistent forest**: classified as forest at every one of the 7 snapshots. The pixel has been continuously forested throughout the full 60-year observation window. High confidence of stable, long-established forest cover.
- **Persistent non-forest**: classified as non-forest at every one of the 7 snapshots. The pixel has never been recorded as forest in 60 years of observation. In the Colombian Llanos context, this is consistent with native savanna, grassland, wetland, or long-established pasture. It does *not* mean the land is degraded; natural savannas are correctly persistent non-forest by design.
- **Non-persistent forest loss**: was forest at some snapshot, non-forest by 2019. This is the primary deforestation signal and the key additionality risk flag for tree-planting projects. Includes losses from 2010-2019.
- **Non-persistent forest gain**: was non-forest at some snapshot, forest by 2019. Captures natural regeneration or plantation establishment.
- **Non-persistent forest**: forest in 1960 and 2019, with non-forest episodes in between. Temporary disturbance with recovery.
- **Non-persistent non-forest**: non-forest in 1960 and 2019, with forest episodes in between. Ephemeral tree cover, possibly transitional or agroforestry.

Important caveat: HILDA+ "non-forest" is a broad class that includes agricultural land, pasture, shrubland, savanna, and wetland; it does not distinguish natural non-forest from human-modified land. High persistent non-forest in this screen is therefore ambiguous without T2 context: in a Grasslands/Savannas biome (T2), it reflects natural baseline; in a Tropical Moist Forest biome (T2), it may indicate long-standing degradation. T2 and T3 must always be interpreted together.

Key Assumptions

Author Assumptions (Estoque et al. 2022, HILDA+)	Our Implementation Choices
The underlying data is HILDA+ (Winkler et al. 2021): a global land use change product at 1 km resolution in the Eckert IV projection (ESRI:54012). It covers 7 decadal snapshots: 1960, 1970, 1980, 1990, 2000, 2010, and 2019.	We reproject the bounding box into the raster's native Eckert IV CRS before clipping. The raster itself is never warped or resampled. The nodata value is 255 and is excluded from all statistics via masked array conventions.
128 unique pixel values encode all possible trajectory sequences across the 7 snapshots. A companion value attribute table (VAT) maps these 128 codes to 6 broad labels: persistent forest, persistent non-forest, persistent forest gain, persistent forest loss, non-persistent forest, and non-persistent non-forest. This mapping is embedded in the published dataset.	We read the VAT dynamically from the ZIP archive using the same decoding logic applied to the Griscom raster. Category percentages are aggregated from all pixel values belonging to each label, not from raw pixel value comparisons.
The category "non-persistent forest loss" captures all pixels where forest was recorded at some point between 1960 and 2019 and non-forest was recorded by 2019. This includes relatively recent losses from 2010 to 2019. Uncertainty is highest in cropland and pasture-dominated areas where transitions are frequent. The 1 km resolution means that sub-kilometre scale deforestation events may not be captured.	All 8 trajectory categories are reported without further aggregation at this stage. Persistent non-forest is flagged when it exceeds 60% of the project area, as high persistent non-forest cover in the context of a planned tree-planting intervention warrants additional scrutiny of intervention design and additionality claims.

Guardrails

Check	Threshold	Result	Status
Valid pixel coverage within bounding box	Above 80%	111 valid pixels (96.1% of clipped extent)	PASS
Category percentages sum to 100%	Within 2% of 100%	100.00%	PASS

Results

Forest Cover Trajectory (Estoque et al. 2022)	Coverage	Distribution
Persistent non-forest	100.0%	

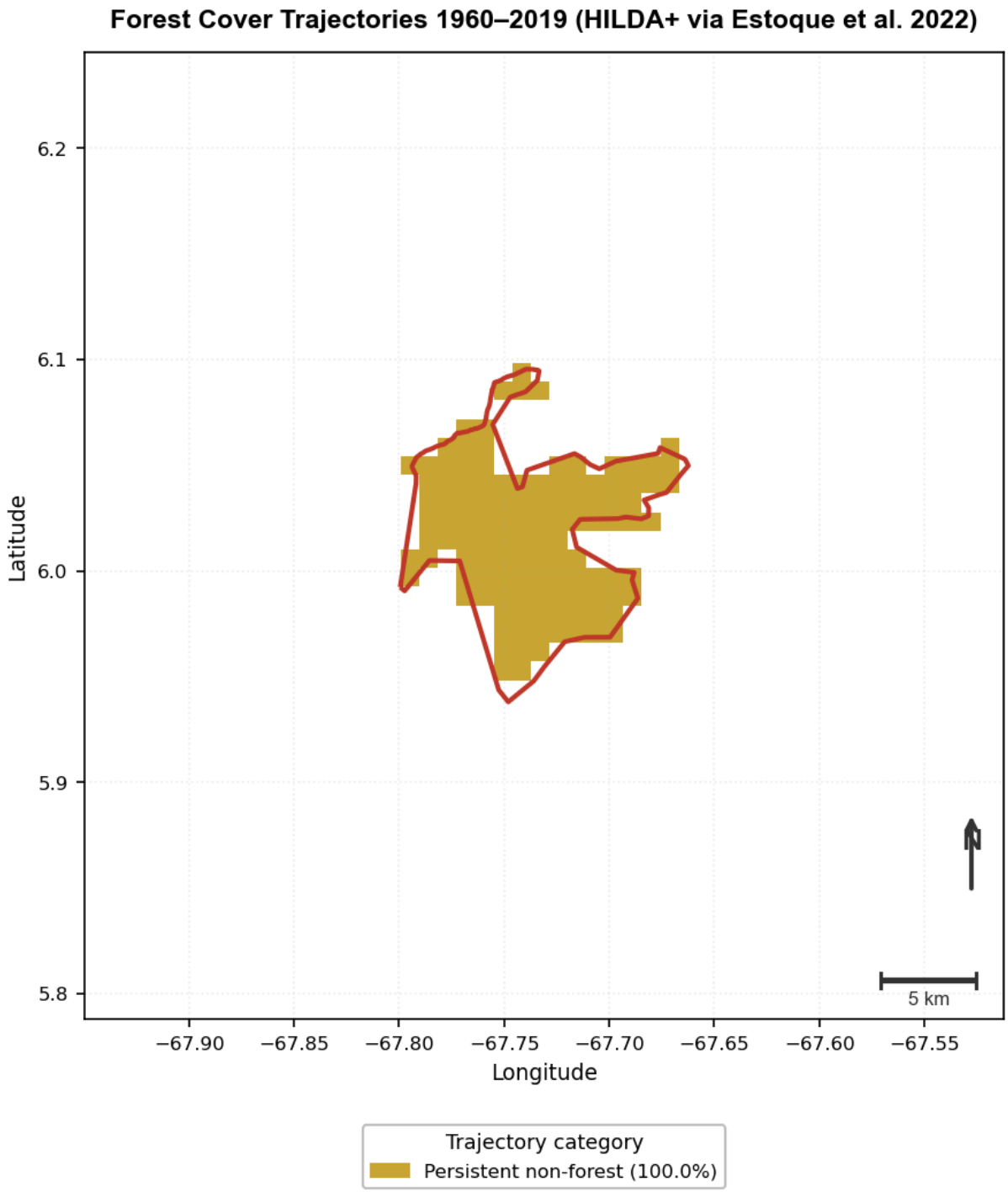


Figure 1: Forest cover trajectories, 1960-2019. Left: spatial choropleth of HILDA+ trajectory class assigned to each 1 km pixel. Right: area breakdown by trajectory class. Source: Estoque et al. 2022 via HILDA+ (Winkler et al. 2021).

T3b - GLC_FCS30D Land Cover Validation (Landsat 30m, 1985-2022)

GLC_FCS30D (Liu et al. 2020) provides 30 m Landsat-derived land cover classifications at 26 time steps from 1985 to 2022. This layer independently validates the T3 (Estoque/HILDA+) persistence finding at 30 times finer resolution and characterises the composition of the forest signal present within the parcel boundary.

Parameter	Value
Dataset	GLC_FCS30D v202106, Google Earth Engine (sat-io open datasets)
Resolution	30 m (Landsat)
Time steps	3 five-year composites (1985, 1990, 1995) + 23 annual maps (2000-2022)
Parcel coverage	Pixel count read from pre-exported CSV; consistent with boundary area at 30 m resolution
Gallery forest proxy	Code 52 (Closed Evergreen Broadleaf Forest); inferred as riparian gallery forest in Llanos biome context. Biome-specific interpretation, not universal.
Collection offset	Step-down at 1995-2000 boundary (1.2 pp) is an inter-product calibration artefact. All primary metrics use post-2000 annual data.
Runtime source	Results read from pre-exported `orojo_colombia_glc_timeseries.csv`; no Earth Engine calls at report generation time.

Results

Metric	Value	Interpretation
Mean grassland % (1985-2022)	89.3%	Dominant cover class across all 26 time steps
Mean forest % (post-2000)	4.3%	Annual product (2000-2022); internally consistent
Forest variability CV (post-2000)	3.5%	Low CV = stable signal; high CV = inter-annual flux
Forest trend (post-2000)	+0.01% per year (stable)	Positive = increasing woody cover; near-zero = stable
Dominant forest class	Code 52 (3.4%)	Most abundant per-class forest column in post-2000 annual record
Gallery forest inferred	Yes	Code 52 >70% of forest signal and CV <10%

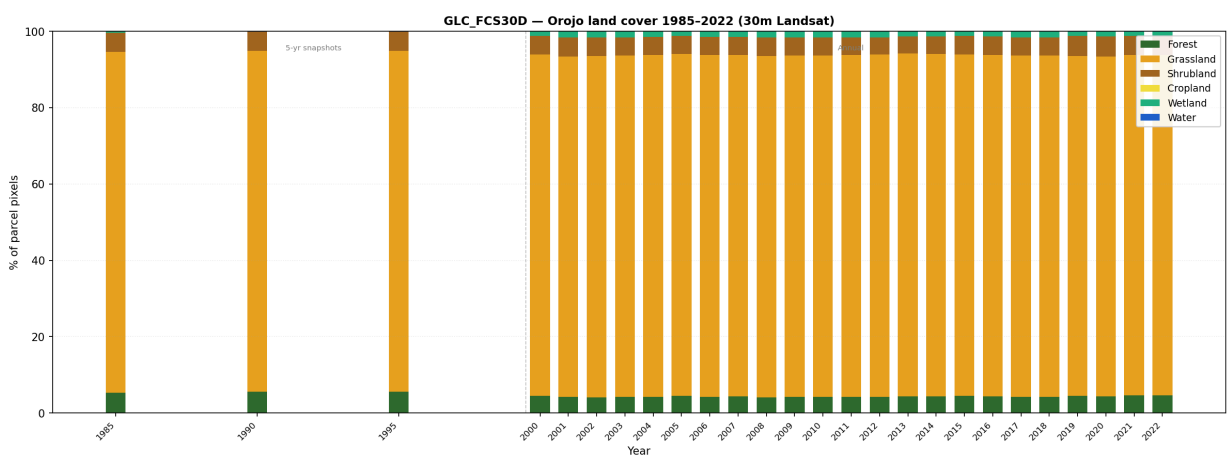


Figure 3b: GLC_FCS30D land cover time series, 1985-2022. Stacked bar chart of land cover group percentages at the project parcel. Grassland (amber) dominates all time steps. Vertical dashed line marks the 5-year to annual product boundary (calibration offset, not an ecological event). Source: GLC_FCS30D v202106, Google Earth Engine (sat-io).

GREEN - GLC_FCS30D confirms persistent grassland dominance.
Mean grassland cover is 89.3% across 26 time steps (1985-2022). The forest signal (4.3% post-2000, CV 3.5%) is stable and dominated by code 52 (Closed Evergreen Broadleaf Forest), consistent with gallery forest (bosque de galería) along riparian corridors in the Llanos biome. No increase in woody cover is detected over the 2000-2022 annual record.

T4 - NCS Reforestation Opportunity (Griscom et al. 2017)

The Griscom NCS reforestation map identifies land that is biophysically capable of supporting forest and where forests were historically present, but where the current cover is non-forest. The map explicitly excludes reforestation in areas "where forests are not the native cover type," citing biodiversity harm from tree planting in native grassy biomes (Methods; citing Veldman et al. 2015, Tyranny of trees in grassy biomes). This safeguard is encoded in the raster itself: pixels in native non-forest biomes carry NCS_Refor = 0 by design. This layer does not assess economic or political feasibility; it is a biophysical filter only.

Key Assumptions

Author Assumptions (Griscom et al. 2017)	Our Implementation Choices
Reforestation opportunity is defined as areas that are currently non-forest, were historically forested, and are biophysically capable of supporting forest. Opportunity here refers to biophysical suitability only. Economic feasibility, land tenure, and political context are outside the scope of this dataset.	We clip the raster to the bounding box and decode all pixel values via the companion VAT DBF file (19,506 entries, columns: Value, Count, NCS_Refor). Pixel values are encoded integers in the 10,000-range and must be decoded through the VAT. Direct comparison to raw integer values produces incorrect results and was a bug in an earlier version of this code.
The map explicitly excludes reforestation in areas "where forests are not the native cover type," citing biodiversity harm from tree planting in native grassy biomes (Methods; citing Veldman et al. 2015, Tyranny of trees in grassy biomes). This safeguard is encoded in the raster itself: pixels in native non-forest biomes carry NCS_Refor = 0 by design.	RESOLVE biome names are used as a separate contextualisation layer alongside this result. When the dominant biome is a grassland or savanna type, low NCS overlap is noted as an expected outcome of the Griscom safeguard rather than a data problem. The principle is Griscom's; the mapping of RESOLVE biome names to interpret the result is our implementation choice.
Raster resolution is approximately 739 m in the World Cylindrical Equal Area projection (ESRI:54034). The dataset is a global-scale product and should not be used as the sole basis for site-level reforestation suitability. Local assessments are required to confirm biophysical conditions at project scale.	We reproject the bounding box into the raster's native ESRI:54034 CRS before clipping. 0% of pixels in the Orojo bounding box were absent from the VAT, confirming full decode coverage for this site.

Guardrails

Check	Threshold	Result	Status
NCS reforestation overlap	Below 10% = FLAG; in savanna or grassland biomes, low overlap is expected per Griscom safeguard and is noted in context	0.00% (Tropical & Subtropical Grasslands, Savannas & Shrublands)	FLAG (expected)
VAT decode coverage (unmapped pixels)	Below 1%	0.00%	PASS

Results

Metric	Value
Valid pixels in bounding box	200
NCS-eligible pixels (NCS_Refor = 1 per VAT)	0
NCS reforestation overlap	0.00%
Dominant biome context (from T2)	Tropical & Subtropical Grasslands, Savannas & Shrublands

Expected outcome: dominant biome (Tropical & Subtropical Grasslands, Savannas & Shrublands) is excluded by the Griscom safeguard. Low NCS overlap reflects the dataset design, not a data problem.

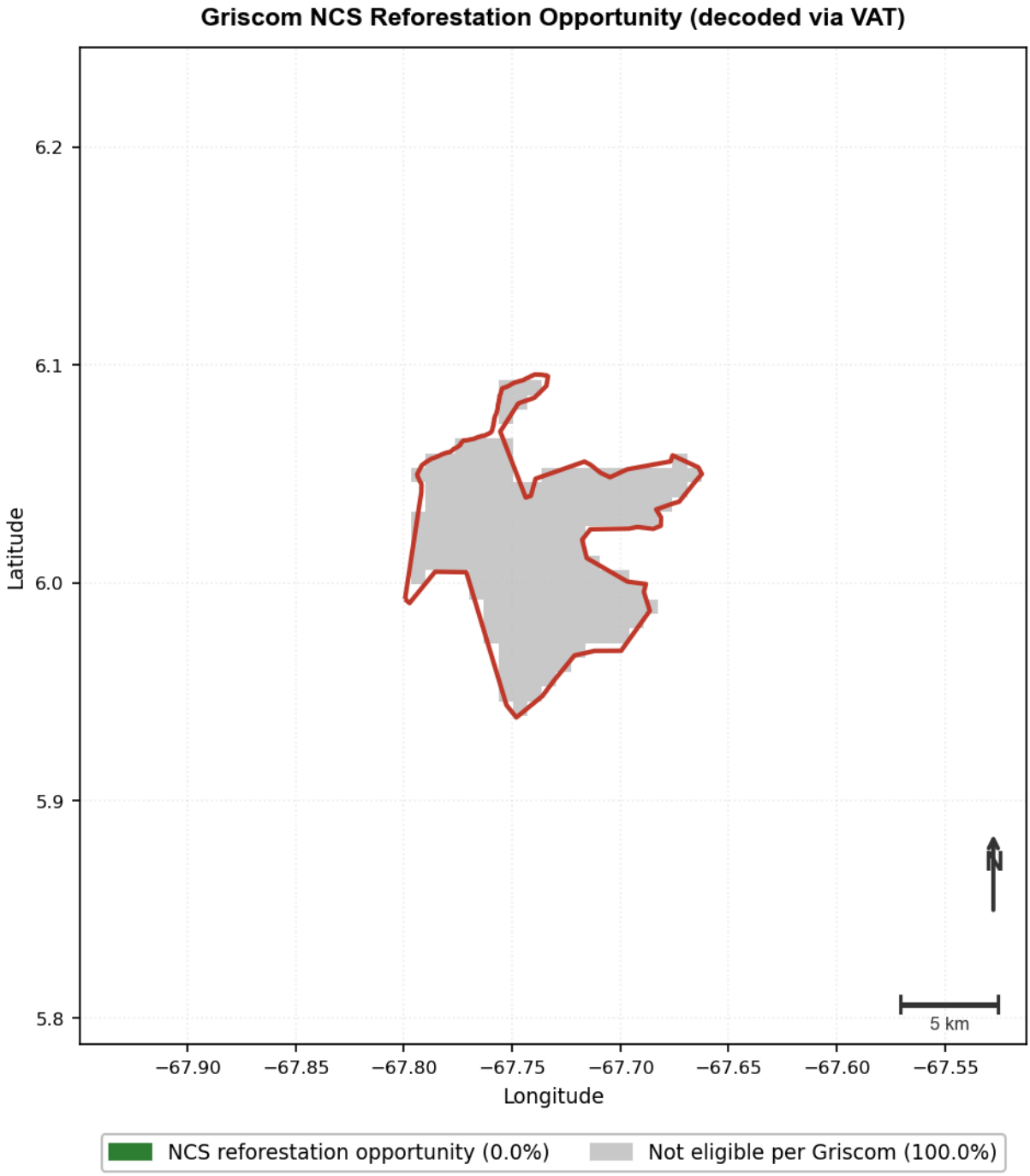


Figure 1: NCS reforestation opportunity map. Griscom et al. 2017 eligibility raster clipped to project bounding box. Green pixels: NCS-eligible (NCS_Refor = 1 per VAT). Grey: ineligible or excluded native non-forest biome. Source: Griscom et al. 2017, PNAS.

T5 - Albedo Climate Implications (Hasler et al. 2024)

Land cover change alters how much sunlight a surface reflects (its albedo). In some landscapes, this biophysical effect can partially or fully offset the carbon sequestration benefit of reforestation. The Hasler dataset quantifies this albedo offset as a percentage of the carbon benefit: a value of -50% means the albedo warming effect would halve the net climate benefit, and a value of -100% means the intervention would be net climate-negative despite its carbon uptake. This layer is critical for understanding whether a given intervention is truly climate-positive when all radiative effects are considered.

Key Assumptions

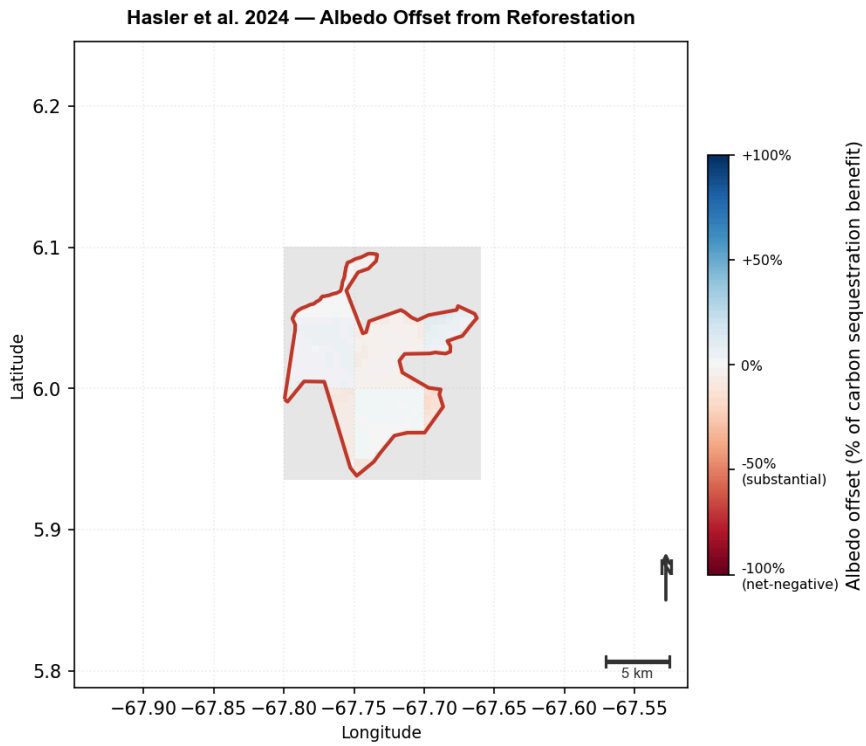
Author Assumptions (Hasler et al. 2024)	Our Implementation Choices
Values represent the albedo radiative forcing offset expressed as a percentage of the carbon sequestration benefit for the same land cover transition. Positive values indicate a cooling co-benefit from albedo change; negative values indicate a warming penalty that reduces the net climate value of the intervention.	We use the composite file, which assigns each pixel the most likely open-to-vegetation transition type based on the surrounding landscape context. This is the appropriate choice for a screening-level assessment where the specific intervention type has not yet been decided. Project-specific analysis should use the single-transition files (e.g., GRA2FOR, GRA2SAV) once the intervention pathway is defined.
The composite map is not transition-specific. Pixel values reflect the expected dominant transition for that landscape context, not a user-specified intervention. Uncertainty is highest in heterogeneous landscapes near biome boundaries where multiple transition types are plausible.	Pixels with absolute values at or above 10,000 are treated as undefined and excluded from all statistics. This follows the Hasler paper convention for flagging extreme or undefined transition scenarios. The raster carries no formal nodata attribute, so this cap-based masking is the primary nodata filter.
The dataset covers global land areas at 0.005-degree resolution (approximately 500 m at the equator) in EPSG:4326. The analysis accounts for radiative forcing from albedo change but does not include other biophysical effects such as evapotranspiration changes or surface roughness effects, which can also influence local and regional climate.	Statistics are computed over all valid (non-capped) pixels within the bounding box. We report the median offset, the percent of pixels in each severity bucket, and whether any net climate-negative pixels are present. The spatial map uses a diverging blue-red colormap (blue = cooling, red = warming) clipped at +/-100% for display; values between +/-100% and +/-10,000 are still included in statistics but appear at the color extreme.

Guardrails

Check	Threshold	Result	Status
Median albedo offset	Below -50% triggers a substantial offset flag (RED); Above 0% is a positive co-benefit	+0.7%	POSITIVE
Net climate-negative pixels (below -100%)	Any pixels present triggers an AMBER flag	0.0% of pixels	PASS
Cap pixels excluded as undefined	Below 5% expected for tropical sites	0.0%	PASS

Results

Metric	Value
Transition type used	composite (most likely open-to-vegetation transition per Hasler et al. 2024)
Median albedo offset	+0.7%
Pixels with net cooling co-benefit (above 0%)	61.4%
Pixels with warming offset (0 to -50%)	38.6%
Pixels with substantial warming offset (-50 to -100%)	0.0%
Net climate-negative pixels (below -100%)	0.0%
Cap pixels excluded from statistics	0.0%



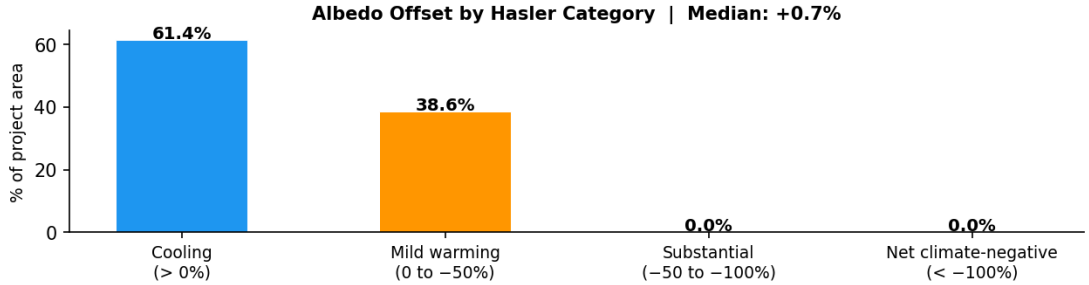


Figure 1: Albedo offset spatial distribution. Left: pixel-level albedo offset (%; blue = cooling co-benefit, red = warming penalty, display clipped at ±100%). Centre: distribution of offset values across valid pixels. Right: pixel count by severity bucket. Source: Hasler et al. 2024, Nature Communications.

T6 - IPLC Land Proximity Screen (LandMark Global Platform v202509)

This screen checks whether the project parcel overlaps with or is within 20 km of any mapped Indigenous Peoples' or Local Community (IPLC) land as recorded in the LandMark Global Platform (September 2025 release). Two polygon layers are screened: the IPLC layer (124,616 territories with documented or customary-tenure status, sourced from national registries and indigenous-data partnerships including ANT Colombia and Wataniba/RAISG) and the Indicative layer (17,539 estimated or undemarcated territories); distance values are nearest-edge, not centroid-to-centroid.

Flag thresholds:

- **RED**: direct overlap with any IPLC or Indicative polygon. FPIC required under ILO Convention 169 / Colombian Law 21/1991, regardless of overlap area.
- **AMBER**: nearest IPLC boundary within 20 km, no direct overlap.
- **GREEN**: no IPLC features within 20 km.

Key Assumptions

Data source and dataset assumptions (LandMark v202509)	Our implementation choices
LandMark does not claim to be exhaustive. Oral traditions, seasonal use areas, and territories not yet mapped or submitted to the platform will not appear. Absence of a feature in LandMark confirms only the absence of a mapped record at this dataset version.	We report this screen as a minimum floor, not a ceiling. If the developer holds local stakeholder information indicating additional communities, that information should be layered on top of this screen in the ESDD.
The `area_gis` field in the IPLC layer stores territory area in hectares. This was verified by comparing stored values against UTM-computed polygon geometry area for known reference features. Small discrepancies are attributable to boundary version differences between the LandMark polygon and other cadastral records.	We display `area_gis` as-is in hectares. We do not recompute territory areas; territory area reporting is secondary to overlap and distance classification.
The 20 km buffer is a precautionary standard screening distance, consistent with IFC Performance Standard 7 guidance and Equator Principles EP4. It is not a legal threshold; project-specific ESDD may warrant a different zone based on activity type, topography, and hydrology.	We apply the 20 km default consistently. The buffer is constructed in UTM metric space and reprojected to EPSG:4326 for display. The same UTM zone (EPSG:32619) is used throughout T1-T6 for spatial consistency.

Guardrails

Check	Threshold	Result	Status
Direct overlap with IPLC or Indicative territory	Any overlap = RED flag	1 direct overlap	RED
IPLC features within 20 km buffer (no overlap)	Any within buffer = AMBER flag	7 features	N/A

Results

Parameter	Value
Parcel area (UTM Zone 19N)	110.9 km²
Buffer applied	20 km
IPLC layer features in search envelope	10
Indicative layer features in search envelope	0
Direct overlaps	1
Features within 20 km (no overlap)	7
Screen result	RED

RED - direct overlap with mapped IPLC territory.
The parcel intersects 1 mapped IPLC territory. Under ILO Convention 169 (ratified as Colombian Law 21/1991), any intersection with a Resguardo Indígena triggers the Free, Prior and Informed Consent (FPIC) requirement regardless of overlap area. 7 additional territories lie within the 20 km proximity buffer.

Proximity Table: All Features

Layer	Territory name	ISO	Category	Doc. status	Territory area	Distance / overlap
IPLC	**Resguardo Indígena Guahibo de Guacamayas y Maipore**	COL	Resguardo Indígena	Documented	16,685 ha	0.32 km² (0.29% of parcel)
IPLC	**Resguardo Indígena Guahibo de Caño Bachaco**	COL	Resguardo Indígena	Documented	6,303 ha	3.7 km
IPLC	**Resguardo Indígena Guahibo de Caño Guaripa**	COL	Resguardo Indígena	Documented	8,137 ha	9.7 km
IPLC	**Resguardo Indígena Wasapana Dagua**	COL	Resguardo Indígena	Documented	5,877 ha	10.7 km
IPLC	**Resguardo Indígena Guahibo de Caño La Hormiga**	COL	Resguardo Indígena	Documented	4,134 ha	11.4 km
IPLC	**Resguardo Indígena Guahibo-Amorua Caño Mesetas Dagua y Murcielago**	COL	Resguardo Indígena	Documented	82,687 ha	11.7 km
IPLC	**Resguardo Indígena Kanalitojo**	COL	Resguardo Indígena	Documented	1,300 ha	17.5 km
IPLC	**Territorio tradicional Jivi**	VEN	Área Indígena de uso tradicional	Held or used under customary tenure	57,581 ha	19.1 km
IPLC	**Territorio tradicional Jivi**	VEN	Área Indígena de uso tradicional	Held or used under customary tenure	67,537 ha	21.1 km *(advisory)*
IPLC	**Territorio tradicional Uwottüja**	VEN	Área Indígena de uso tradicional	Held or used under customary tenure	2,285,502 ha	28.1 km *(advisory)*

Red background: direct overlap. Amber: within 20 km buffer. Grey (advisory): beyond buffer threshold. Source: LandMark Global Platform v202509, Rights and Resources Initiative.

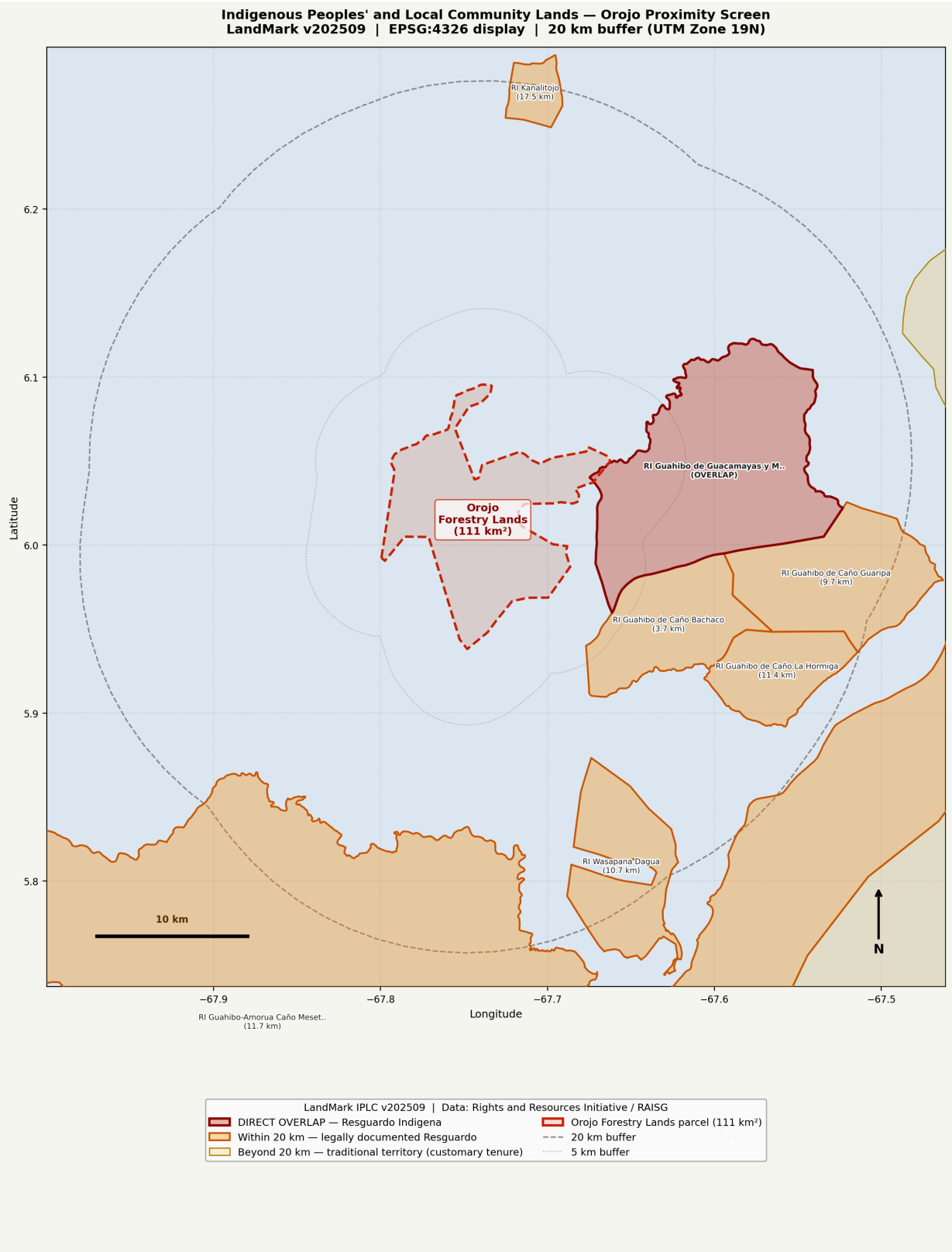


Figure 1: IPLC proximity map. Project parcel boundary (red), 20 km buffer ring (dashed), and all IPLC features within or near the search envelope. Red fill: direct overlap. Amber fill: within 20 km. Source: LandMark Global Platform v202509.

Generated by the Orojo Geospatial Screening Pipeline | Layers T1 through T6 | 18 March 2026

Data sources: RESOLVE Ecoregions (Dinerstein et al. 2017, One Earth), HILDA+ via Estoque et al. 2022 (Environmental Research Letters), Griscom et al. 2017 Natural Climate Solutions (PNAS), Hasler et al. 2024 Albedo Offset (Nature Communications), LandMark Global Platform v202509 (Rights and Resources Initiative).

This report is a screening-level assessment. All results are based on global-scale datasets at resolutions of 500 m to 1 km. They should be interpreted alongside site visits, local ecological knowledge, and project-specific measurements before any investment or management decisions are made.